

Failures of Technology, Whistleblowing, Disclosure, and Unintended Consequences

ENGR49580D Technology, Society and Sustainability

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- Unintended Consequences and Failures of Technology
 - ‘Lighter than Air’ Ships
 - US Space Shuttle
 - DDT
- Whistleblowing and Disclosure

Failures of Technology

Background

- The consequences of these technologies (or their failures) may not be foreseen
- Impacts may reach beyond anticipated consequences
 - May cause widespread environmental impacts
 - May change public perceptions of a technology or industry

Failures of Technology

Hindenburg — Background

- In the inter-war years (between World Wars I and II), ‘Lighter than Air’ Ships were a popular mode of international travel
- On ship amenities were comparable to ocean liners / cruise ships
 - Private Cabins
 - Lounges
 - Smoking Room
 - Dining Room

Failures of Technology

Hindenburg — Background

- The largest commercial airship was the German *Hindenburg*
- The Hindenburg's full compliment was 50 passengers and 40 crew

Failures of Technology

Hindenburg

- Of 97 people onboard, 35 were killed in the explosion
- News of the tragedy spread across the globe quickly through newspapers and newsreels
- The horror of the fire and explosion lead quickly to the end of the 'Lighter than Air' Ship industry

Failures of Technology

Hindenburg — Causes

- The source of ignition is not know for certain
- The Hindenburg was filled with hydrogen, not the preferred helium
 - At the time helium extraction was limited to the US
 - Due to post World War I sanctions, the US was not exporting helium to Germany
 - Hydrogen, was able to be produced domestically in Germany

Failures of Technology

Space Shuttles Challenger and Columbia — Background

- The Space Transportation System (STS) program was NASA's primary vehicle for moving heavy payloads and personnel into orbit
 - Commonly known as the Space Shuttle
 - 5 Shuttles flew 135 missions from 1981 to 2011
 - Over the life of the program, a number of concerns were raised about the space shuttle's technology

Failures of Technology

Space Shuttle Challenger — Background

- 28 January 1986 STS-51-L
- Regularly scheduled shuttle mission
- Was to transport first teacher into space
 - Due to this event, significant media attention was paid to the launch

Failures of Technology

Space Shuttle Challenger — Cause

- O-rings joining sections of the solid rocket boosters failed causing a fuel leak that ignited the external fuel tank
 - O-ring failure was caused by low temperature
- Engineers were aware of the potential for O-ring failure in low temperatures
 - The organizational design at NASA made it difficult for them to communicate with senior administrators
 - NASA administration was under pressure to proceed with the launch that had been delayed 4 times already

Failures of Technology

Space Shuttle Challenger — Consequences

- None of the 7 crew members survived
- It is believed that all crew survived the explosion, however they would have been trapped in the shuttle cabin which had no escape mechanism
- After the Challenger disaster there was a 32 month suspension of shuttle flights

Failures of Technology

Space Shuttles Columbia — Background

- 16 January 2003 STS-113
- Regularly Scheduled Shuttle Mission

Failures of Technology

Space Shuttles Columbia — Cause

- During launch insulating foam fell from the main fuel tank damaging heat resistant tiles on the left wing
 - Flight and ground crews were aware of the lost insulation, but not of the damage to heat tiles
- Without the heat resistant tiles, the shuttle could not survive the heat of reentry
- The loss of insulating foam on launch was common and had not resulted in damage in the past
 - Insulation loss incidents were regularly ignored despite regulations

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Space Shuttles Columbia — Consequences

- None of the 7 crew survived
- Shuttle program was on hiatus for 29 months
 - This was at the end of the construction of the ISS

Failures of Technology

DDT — Background

- First Synthesized 1874
- Recognized as a pesticide 1939
 - Identified as particularly effective in controlling Malaria / Typhoid Mosquitos
 - Also useful in controlling agricultural pests

Failures of Technology

DDT — Background

- In 1962 the book *Silent Spring* identified the use of DDT as contributing to the large-scale die off of several bird species including the bald eagle and California condor
 - Exposure to DDT caused thinning of eggshells and inviable eggs
- DDT is a bio accumulator
 - It accumulated in the tissues of small mammals that eat sprayed crops
 - It further accumulated in predatory birds
 - California condors were particularly vulnerable as scavengers

Failures of Technology

DDT — Consequences

- DDT banned in the US in 1972
- DDT is now regulated as a persistent organic pollutant (POP)

Whistleblowing and Disclosure

Definition

- Whistleblowing refers to the disclosure of organizational behaviour that may be illegal, unethical, unsafe, or fraudulent with the intention of the disclosure resulting in organizational change
- Whistleblowing may be:
 - Internal, typically through anonymous reporting channels including 3rd party service providers
 - External, to oversight agencies such as law enforcement, or regulators
 - External, to the media

Whistleblowing and Disclosure

Typical Whistleblower

- The typical whistleblower has:
 - a long history of successful employment
 - high ethical standards / strong religious beliefs
 - the belief that whistleblowing will correct the problem

Whistleblowing and Disclosure

Consequences

- Employment Consequences:
 - Job action (termination, suspension, demotion, pay / hours reduction)
 - Blacklisting (repetitional damage, poor references)
 - Workplace bullying
- Criminal
 - Charges of espionage / treason

Whistleblowing and Disclosure

Consequences

- What if the whistle blower is wrong?
- Rushed investigations initiated by whistleblowers may lead to unjustified accusations and repetitional harm or other consequences
 - Health Canada 2012 whistleblower

Whistleblowing and Disclosure

Whistleblower Protection

- Public Sector (Canada)
- Public Service Integrity Commissioner
 - Provides a reporting mechanism as well as protection for whistleblowers
 - Authority to investigate and requirement to report

Whistleblowing and Disclosure

Whistleblower Protection

- Private Sector (Canada)
- Governed by the *Competition Act*
 - Applies only to criminal offences under the act reported to the competition bureau or law enforcement
 - Reporter's anonymity is protected
 - Prohibition against employer reprisal
- Similar protections for employees under the criminal code